

Adding a central router to network topology

09/20/2026 01:05:04

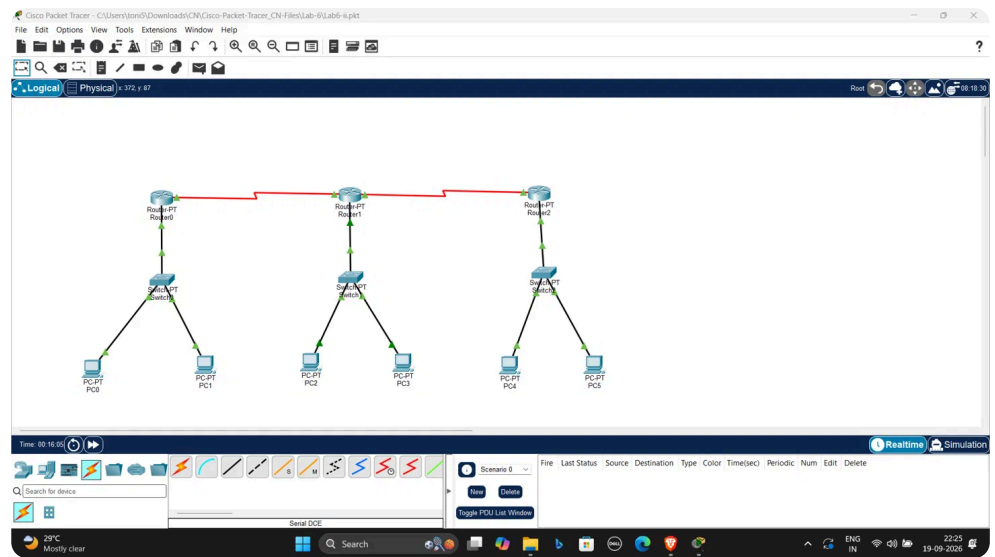
<https://claude.ai/chat/1892d683-cdbe-4c1e-9023-d4a5591a4ae3>

[Lab6-iiia.png]

Your faculty most likely wants a fourth router (Router3) in the middle,...

So what shall be the fastethernet0/0 for router 3

Use a new network that isn't used anywhere else. Networks 10 to 80 are...



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Working out new serial modules and subnet addressing. ▾

- ⌚ Working out new serial modules and subnet addressing.
- ✓ Done

Your faculty most likely wants a fourth router (Router3) in the middle, linked to Router0, Router1 and Router2. Here is how to add it to your current setup.

Phase 1: Add the router and serial modules

- Place Router3.** Drag a **Router-PT** into the empty space above and between your three routers, so it sits centrally.
- Add serial modules where needed.** Each new link needs a free serial port on both ends:
 - Router3** needs 3 serial ports.
 - Router0, Router1 and Router2** each need 1 more.
 - For each router: click it, open the **Physical** tab, click the **power switch** to turn it off, and drag a serial module (e.g. **PT-ROUTER-NM-1S**) from the left list into an empty slot. Then turn the power back on.
 - Note the new port names (e.g. **Serial3/0**, **Serial4/0**), because yours may differ from the ones below.

Phase 2: Cable the new links

Use the **Serial DCE** cable. Click the outer router first, so it becomes the DCE side and gets the clock rate.

- **Router0** new serial → **Router3** **Serial2/0**
- **Router1** new serial → **Router3** **Serial3/0**
- **Router2** new serial → **Router3** **Serial4/0**

Your existing Router0–Router1 and Router1–Router2 links stay as they are. The lines will show red until the interfaces are configured.

Phase 3: Configure the interfaces

Use the **Config** tab for each router, and turn **Port Status: On** for every new port.

Router	Interface	IP Address	Mask	Clock Rate
Router0	new serial	60.60.60.1	255.0.0.0	2000000
Router1	new serial	70.70.70.1	255.0.0.0	2000000
Router2	new serial	80.80.80.1	255.0.0.0	2000000
Router3	link to Router0	60.60.60.2	255.0.0.0	none
Router3	link to Router1	70.70.70.2	255.0.0.0	none
Router3	link to Router2	80.80.80.2	255.0.0.0	none

If a link stays red, check which end is the DCE. The clock rate must be set on that end.

Phase 4: Extend RIP

Open **Config** → **RIP** on each router and add the new networks:

- **Router0:** add 60.0.0.0
- **Router1:** add 70.0.0.0
- **Router2:** add 80.0.0.0
- **Router3:** add 60.0.0.0, 70.0.0.0 and 80.0.0.0

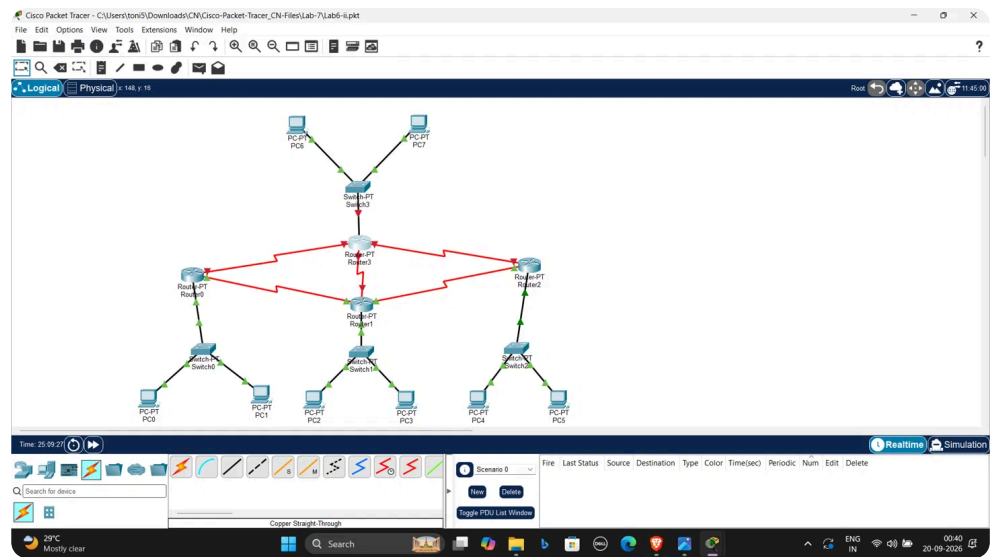
The existing entries on Router0, Router1 and Router2 (10, 20, 30, 40 and 50) stay unchanged.

Phase 5: Verify

1. On **PC0**, run `ping 30.30.30.1` (PC4). The first packet may time out, but the rest should succeed.
2. Run `tracert 30.30.30.1` from PC0. RIP picks the path with the fewest hops, so the route may go directly through Router3 rather than through Router1.
3. Ping Router3's interfaces (e.g. `ping 60.60.60.2`) from any PC to confirm it is reachable.
4. To demonstrate that the central router provides redundancy, delete the Router0–Router1 cable. Pings from PC0 to PC2 should still work through Router3 after RIP reconverges, which can take a minute.

If you want a pure star topology, delete the direct Router0–Router1 and Router1–Router2 cables and the ports for `40.x` and `50.x`. Ask your faculty which they expect, since a "central" node often means a star layout.

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So what shall be the fastethernet0/0 for router 3

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Working out the addressing scheme for a router interface. ▾

- ⌚ Working out the addressing scheme for a router interface.
- ✓ Done

Use a new network that isn't used anywhere else. Networks 10 to 80 are already taken, so **90.x.x.x** is the next free one.

Router3 → Config → FastEthernet0/0

- IPv4 Address: **90.90.90.1**
- Subnet Mask: **255.0.0.0**
- Port Status: **On** (the red arrow at the Switch3 end shows this port is still down)

PC6

- IP: **90.90.90.2**
- Mask: **255.0.0.0**
- Gateway: **90.90.90.1**

PC7

- IP: **90.90.90.3**
- Mask: **255.0.0.0**
- Gateway: **90.90.90.1**

RIP on Router3 (Config → RIP): add **90.0.0.0** along with the **60.0.0.0**, **70.0.0.0** and **80.0.0.0** you already added. The other routers don't need any change, because RIP will pick up **90.0.0.0** from Router3.

To test, ping from PC6 to PC0 (**10.10.10.1**) and PC4 (**30.30.30.1**). The first packet may time out, but the rest should succeed.

Your other routers use **.0** addresses (like **10.10.10.0**) for the gateway, and that also works in Packet Tracer with a **/8** mask. If you want Router3 to match that style, use **90.90.90.0** instead, but keep the PC gateways the same as whatever you set on the router.

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